CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in Large Language Models, AI agents and AI-assisted software development with Claude Code, with hands-on Docker, Kubernetes, CI/CD and Infrastructure as Code. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by cloud deployment on AWS and Azure across enterprise and start-up environments.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present | Muldestausee, Germany

- Owned the knowledge-extraction pipeline end-to-end as sole developer, from data ingestion and chunking to prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Designed and optimized the multi-agent orchestration in LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel) for complex regulatory and technical documentation, raising precision/recall from 33%/45% to 93%/90% measured against an SAP-system-derived ground truth.
- Engineered a deterministic, code-based extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing accuracy toward 100%.
- Designed and implemented a Model Context Protocol (MCP) service that exposes the knowledge graph for millisecond-latency retrieval, integrated into Claude Code for agentic development workflows.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) on Azure where domain experts validate and approve knowledge-graph changes.
- **Tech stack: Kubernetes, Docker, Azure, Claude Code, LiteLLM, MCP, Python, LangGraph, LangChain, Neo4j, GraphRAG, RAG, FastAPI, MongoDB, Postgres, React**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 | Leverkusen, Germany

- Designed and deployed an agentic AI application with LangGraph that routes decision-making across vector database queries, general inference and summarization.
- Built the Python/FastAPI backend and a React frontend, tested the APIs with Postman, containerized the stack with Docker and deployed it to AWS with CDK; versioned in GitLab with CI/CD pipelines.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD, Postman, React**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 | Leverkusen, Germany

- Automated evals of RAG generation and retrieval with RAGAS and PytrecEval, against chemist-provided ground truth, with serverless AWS infrastructure for PrismaChat RAG evaluation (Lambda, SQS, DynamoDB) built with CDK.
- Benchmarked multilingual embedding models on over 1M chunks of internal chemical data and designed RAG pipelines (chunking, vector databases, indexing, fine-tuning models for retrieval).
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, PyTorch, AWS (Lambda, SQS, DynamoDB, CDK), Azure AI Search, Docker**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes from time-series process data, and delivered a Streamlit dashboard for chemists to simulate parameter changes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for EV load time-series forecasting; the LSTM reached 96% test accuracy with strong performance on unseen data.

PROJECTS

Project: AI Gateway Platform | LiteLLM | Kubernetes | Flux | Vault | Grafana

- Self-hosted AI gateway (LiteLLM proxy) as a shared platform service for development teams: virtual keys per team, token budgets, retries and fallback between LLM providers, and logging of latency, token usage and cost for every request; teams onboard through a configuration file.
- Runs on a Kubernetes cluster deployed with Flux (GitOps) from a GitHub repository via a Helm chart, with secrets from HashiCorp Vault and Grafana dashboards over Loki logs and Prometheus metrics; a GitHub Actions pipeline runs tests and an image scan on every change.

Project: Spec-Driven Coding Agent Harness | Claude Code | MCP | Langfuse | GitHub Actions

- Harness that turns a plain-language feature request into testable specifications (acceptance criteria written as pytest tests) and lets Claude Code implement them under a rule set: project conventions, custom skills, and hooks that block edits to protected files and secrets; an MCP server gives the agent the spec and task context.
- A GitHub Actions CI/CD pipeline gates every agent-written change with automated tests, static security analysis and dependency scanning, plus an agent evaluation suite over benchmark tasks (pass rate, cost, turns per task); traces of the agent runs go to Langfuse.

TECHNICAL SKILLSET

- **Cloud & DevOps:** Docker, Kubernetes, CI/CD (GitLab CI/CD), Infrastructure as Code (AWS CDK), Azure, AWS (EC2, Lambda, SQS, DynamoDB, S3, VPC), Git
- **AI-Assisted Development:** Claude Code (custom skills, MCP integration, agentic coding workflows)
- **LLMs & AI Agents:** LiteLLM, LangGraph, LangChain, MCP, RAG, GraphRAG, prompt engineering, evals (RAGAS, PytrecEval), Hugging Face Transformers
- **Programming & Backend:** Python, FastAPI, Python microservices, SQL, Postman, React (beginner)
- **Data & ML:** Neo4j, MongoDB, Postgres, vector databases (Qdrant, Pinecone, Azure AI Search), PyTorch, Scikit-Learn

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
Minor in Computer Science & Engineering
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)